Formelsammlung: Fourier und Statistik

Fourierreihen

Grundlagen

- cos = gerade
- sin = ungerade
- $\omega_0 = \frac{2\pi}{T}$

$$FR\{y(t)\} = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$a_n = \frac{2}{T}\int_0^T y(t)\cos n\omega_0 t dt$$

$$FR\{y(t)\} = \sum_{n=-\infty}^{\infty}(c_n e^{in\omega_0 t})$$

$$a_0 = \int_0^T y(t)$$

$$b_n = \frac{2}{T}\int_0^T y(t)\sin n\omega_0 t dt$$

Fourier transform

$$y(t) \circ - - - - \bullet Y(\omega)$$

Spektralfunktion

$$Y(\omega) = \int_{t=-\infty}^{\infty} y(t)\cdot e^{-i\omega t}dt$$

Zeitfunktion

$$y(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} Y(\omega)e^{i\omega t}d\omega$$